

JEE MAIN REVISION MASTER: SEMICONDUCTOR DEVICES AND LOGIC GATES

Subject: Physics | Target Scope: JEE Main Previous Years | Empirical Base: 32 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	p-n Junction Diode & Zener Diode Regulation	25 PYQs	Semiconductor Diodes
Moderate	Logic Gates & Truth Tables	7 PYQs	Digital Electronics

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
p-n Junction Diode & Zener Diode Regulation	$\text{Efficiency}_{\text{half-wave}} = 40.6\%$, $\text{Efficiency}_{\text{full-wave}} = 81.2\%$, $V_L = V_Z = \text{const}$, $I_S = I_Z + I_L$
Logic Gates & Truth Tables	$\text{NAND: } Y = \overline{A \cdot B}$, $\text{NOR: } Y = \overline{A + B}$, $\overline{A \cdot B} = \overline{A} + \overline{B}$, $\overline{A + B} = \overline{A} \cdot \overline{B}$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 p-n Junction Diode & Zener Diode Regulation (Very High Yield — 25 PYQs)

Core Principle: Forward and reverse characteristics of p-n junctions, rectifiers, ripple frequency, and Zener diode shunt voltage stabilization.

■ **Caution / JEE Pitfall:** In full-wave rectifiers, output ripple frequency is TWICE the AC input frequency ($2f_{in}$), whereas for half-wave it is f_{in} .

■ **Speed Trick:** In reverse breakdown, Zener diode maintains constant voltage V_Z across its terminals as long as $I_Z \geq I_{Z,min}$.

Representative Source PYQs:

- [2026 Shift] Q#1 (MCQ): "A wafer of pure germanium crystal has two parts X and Y. The end X is obtained by doping with arsenic and Y with indium. It is connected to ..." [Ans: N/A]
- [2026 Shift] Q#2 (MCQ): "In which of the following figures, diode is reverse biased?..." [Ans: N/A]
- [2026 Shift] Q#3 (MCQ): "An n-type and p-type semiconductor can be obtained by respectively doping pure silicon with..." [Ans: N/A]

3.2 Logic Gates & Truth Tables (Moderate Yield — 7 PYQs)

Core Principle: Universal logic gates (NAND and NOR), De Morgan's theorems, and Boolean circuit reduction.

■ **Caution / JEE Pitfall:** NAND and NOR gates are UNIVERSAL gates: any Boolean function can be constructed using only NAND or only NOR gates.

■ **Speed Trick:** Connect inputs of a NAND or NOR gate together to obtain a NOT gate: $\overline{A \cdot A} = \overline{A}$.

Representative Source PYQs:

- [2023 Shift] Q#4 (MCQ): "A truth table for the given circuit is..." [Ans: C]
- [2022 Shift] Q#3 (MCQ): "Which logic gate is represented by the following combination of logic gates?..." [Ans: N/A]
- [2021 Shift] Q#4 (MCQ): "The circuit given represents which of the logic operations?..." [Ans: N/A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: KCET 2026 (Q#1)

A wafer of pure germanium crystal has two parts X and Y. The end X is obtained by doping with arsenic and Y with indium. It is connected to a battery as shown in the figure. Which of the following statements is correct?

- (A) X is p-type, Y is n-type and the junction is forward biased
- (B) X is n-type, Y is p-type and the junction is forward biased
- (C) X is p-type, Y is n-type and the junction is reverse biased
- (D) X is n-type, Y is p-type and the junction is reverse biased

Correct Answer: None

Solution Analysis: Part X is doped with arsenic. Arsenic is a Group 15 element, so it has 5 valence electrons. When it is added to germanium, it gives extra free electrons. So, part X becomes an n-type semiconductor. Part Y is doped with indium. Indium is a Group 13 element, so it has 3 valence electrons. When it is added to germanium, it creates holes. So, part Y becomes a p-type semiconductor. In the battery symbol, the longer line represents the positive terminal, and the shorter line represents the negative terminal. From the figure, part X is connected to the positive terminal and part Y is connected to the

Benchmark Problem 2: KCET 2023 (Q#4)

A truth table for the given circuit is

- (A) B Y 1 1 1 1 0 1 0 1 0 0 0 1
- (B) B Y 1 1 1 1 0 0 0 1 1 0 0 1
- (C) B Y 1 1 1 1 0 1 0 1 1 0 0 1 [Correct]
- (D) B Y 1 1 1 1 0 1 0 1 1 0 0 0

Correct Answer: C

Solution Analysis: The given logic circuit diagram is Output of the logic gate is $Y = \overline{(\overline{A+B}) \cdot (AB)}$

$(AB) = \overline{A+B} + \overline{AB} = A+B+\overline{A}+\overline{B} = (A+\overline{A})+(B+\overline{B})$ [■ $x+\overline{x}=1$] $=1+1=1$ Thus output is always 1 for all value of inputs A and B. Hence, option (c) is correct.